

MyDesign Portfolio

Element B Initial Template

Element B: Documentation and Analysis of Prior Design Attempts

Prior Design Attempt #1 - This or That Machine

Problem

The library needs a "this or that" machine because they want to have an engaging activity that students can interact with weekly to foster connection. The design should be durable enough, entertaining enough, and should follow the other constraints for being around kids all day.

Some of these constraints include:

- Not being able to fall with one push, so that it stayed in one place well.
- If it does fall, so that it did not break into small pieces like glass would or thin plastics would.

Sketches+descriptions

Sketches of the machine design, including a top-down view and a side view. The sketches show a rectangular box with a handle and a small opening.

original sketch

Decision matrix

zoom in :)

Criteria	Option 1	Option 2	Option 3
Cost	Low	Medium	High
Durability	High	Medium	Low
Engagement	Low	High	Medium
Space	High	Medium	Low

Prototype photos

Two photographs of the prototype machine. The left photo shows the machine on a table, and the right photo shows the machine on the floor.

Testing plan

Right wall off so that the ramp and the mechanisms were seen

Test	Duration	Location	Notes
Test 1: Durability	10 min	Library	Test 1: Durability
Test 2: Engagement	10 min	Library	Test 2: Engagement
Test 3: Space	10 min	Library	Test 3: Space

zoom in

Prototype Video

<https://youtu.be/Vx6PB1clqIQ>

video of the prototype machine in operation

Videos

Testing video

Test 1: Durability

Presentation video

<https://youtu.be/hz7ZpCEvU>

Modifications

- We realized that the connectors we designed do not connect well. We plan to revisit this part of the design and come up with a better solution.
- We want to make the design smaller. We realized there is a lot of wasted space and making the chassis smaller will increase the durability.
- We forgot to include a hole for the micro-usb to usb-a cable to slide through.
- We also would like to make the basket's walls a little higher so that the pom poms didn't fall out when they land.

Prior Design Attempt #2 - This or That Machine

Original Design

The original design was a 3D printed machine with a screen and buttons. It was designed to be used by a librarian to manage a collection of books. The design was based on a 3D model of a bookshelf and a 3D model of a book. The design was based on a 3D model of a bookshelf and a 3D model of a book.

Chart on Feedback

Feedback	Response	Implementation
The design was too complicated.	Simple the design.	Simple the design.
The design was too expensive.	Reduce the cost.	Reduce the cost.
The design was too difficult to use.	Improve the usability.	Improve the usability.

New Prototype Sketches + Photos

The design works about the same as before. Most of the changes were made on the appearance, not on how it functions.

Testing

Notes: Only one of our teammates had enough 3d party participants to conduct tests 2 and 3, so only one did these tests.

Test	Test 1: Usability	Test 2: Usability	Test 3: Usability	Test 4: Usability	Test 5: Usability
Test Summary	Users were able to use the machine to find books. The machine was easy to use and the results were accurate.	Users were able to use the machine to find books. The machine was easy to use and the results were accurate.	Users were able to use the machine to find books. The machine was easy to use and the results were accurate.	Users were able to use the machine to find books. The machine was easy to use and the results were accurate.	Users were able to use the machine to find books. The machine was easy to use and the results were accurate.

Extra Iterations

We took the Librarian's feedback, and reduced the size of stumps from 1500 cubic inches to 341.6 cubic inches.

The cost of our design dropped by about 35%. This was possible not only because we made the whole design smaller but also because we made the walls thinner. The amount of filament dropped by 2000 grams from the original.

We also:

- Redid the way the lid connects so that only the librarian could open it.
- Added a hole for the power cable.
- Changed the design of the funnel to make it easier to install the sensors.
- Redid the connectors so that the device could be assembled without the need of soldering.
- Resized the hole for the screen so that the display fit better.
- Changed the ramp angle by making it less steep, and made the walls of the postpump looker higher so that the pump ports did not fall out when they roll down.
- Changed the pump-holder design to make it easier to slide in the question slips.

Presentation Video

BRH-2 Final Video

While our final iteration solved many of the issues we aimed to address previously in our design philosophy, there are still things we would improve to ensure that it follows our design requirements of ease of usability. With our original design we realized that our circular peg connectors 3-d printed roughly and would not fit into its respective slot without sanding it down or applying excessive force. We recognized that this issue occurred because the circular pegs and slots did not print precisely, causing rough edges and inaccurate sizing. Although in our latest iteration we changed the connectors into squares as opposed to circular pegs to make sure that the parts fit together snugly, we realized that it still did not meet our expectations. Several of our design requirements are in the spirit of ease of use, so we wanted to make sure that our device could be easily disassembled and fixed to follow that philosophy. In the same vein, our reset button holder in our second iteration was not as user friendly as we intended it to be. We needed to use a file to sand down the hole where the button would slide into it so it could fit properly. The holder itself was thin and not well supported and could break easily upon force. Additionally, while the wiring was snug and compact, the space to work with is tight making it difficult to work with. Ultimately, even though our latest iteration has made major headway into making an accessible and user-friendly “this or that” machine, there are still ways to further improve our design to make it adhere better to our original goals and design requirements.